

120 Days of Quantitative Finance

Month 4: Stochastic Calculus & Time Series Modeling

Week 14: SDEs and Black-Scholes

Day 94: Geometric Brownian Motion

1 Introduction

Geometric Brownian Motion (GBM) is the standard model used to describe the evolution of stock prices in continuous time.

It incorporates both deterministic growth and stochastic fluctuations.

2 SDE Representation

GBM is defined by the stochastic differential equation:

$$dS_t = \mu S_t dt + \sigma S_t dW_t$$

where:

- μ is the drift (expected return)
- σ is the volatility
- W_t is Brownian Motion

3 Interpretation

- The drift term $\mu S_t dt$ represents proportional growth

- The diffusion term $\sigma S_t dW_t$ introduces proportional randomness
- Changes scale with the level of S_t

4 Solution using Ito's Lemma

Let:

$$f(S_t) = \ln S_t$$

Applying Ito's Lemma:

$$d(\ln S_t) = \left(\mu - \frac{1}{2}\sigma^2 \right) dt + \sigma dW_t$$

5 Closed-Form Solution

Integrating both sides:

$$\ln S_t = \ln S_0 + \left(\mu - \frac{1}{2}\sigma^2 \right) t + \sigma W_t$$

Exponentiating:

$$S_t = S_0 \exp \left[\left(\mu - \frac{1}{2}\sigma^2 \right) t + \sigma W_t \right]$$

6 Distribution of Prices

$$\ln S_t \sim \mathcal{N} \left(\ln S_0 + \left(\mu - \frac{1}{2}\sigma^2 \right) t, \sigma^2 t \right)$$

Thus:

$$S_t \sim \text{Lognormal}$$

7 Key Properties

- Prices are always positive
- Log-returns are normally distributed

- Variance increases linearly with time

8 Financial Significance

GBM is used in:

- Black-Scholes option pricing
- risk-neutral valuation
- Monte Carlo simulations

9 Limitations

- assumes constant volatility
- no jumps or discontinuities
- ignores fat tails

10 Conclusion

Geometric Brownian Motion provides a mathematically tractable and widely used model for asset prices.

It forms the foundation of modern option pricing theory.